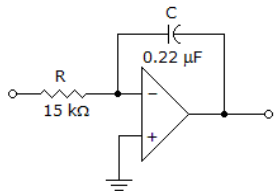


**Multiple Choice:**

1. The slew-rate specification of an op amp is the
 - A. maximum value of positive or negative output voltage.
 - B. maximum rate at which its output voltage can change.
 - C. attenuation against a common mode signal.
 - D. frequency where the voltage gain is one or unity.
2. The output stage of a 741 op-amp is a
 - A. differential amplifier.
 - B. common-base amplifier.
 - C. common-emitter amplifier.
 - D. push-pull amplifier.
3. A steady voltage of -0.75 V is applied to an op amp integrator having component values of $R = 200\text{ k}\Omega$ and $C = 2.5\mu\text{F}$. Assuming that the initial capacitor charge is zero, determine the value of the output voltage 100 ms after application of the input.
 - A. 0.35 V
 - B. 0.15 V
 - C. 0.25 V
 - D. 0.015 V
4. The output of a Schmitt trigger is a
 - A. pulse waveform.
 - B. sawtooth waveform.
 - C. sinusoidal waveform.
 - D. triangle waveform.
5. In an inverting amplifier the input and output voltages are
 - A. in phase.
 - B. 90° out of phase.
 - C. 180° out of phase.
 - D. 360° out of phase.
6. When the same signal is applied to both inverting and non-inverting input terminals of an ideal op-amp, the output voltage would be
 - A. zero (0) V
 - B. $+V_{\text{SAT}}$
 - C. $-V_{\text{SAT}}$
 - D. offset voltage
7. The circuit at the input stage of operational amplifiers
 - A. differential amplifier
 - B. cascaded amplifier
 - C. current mirror
 - D. complementary amplifier
8. A differential amplifier has an open-loop voltage gain of 120. The input signals are 2.45 V and 2.35 V . Calculate the output voltage of the amplifier.
 - A. 12.0 V
 - B. 12.5 V
 - C. 11.5 V
 - D. 10.0 V
9. _____ is a mathematical process for determining the rate of change of a function.
 - A. Integration
 - B. Differentiation
 - C. Summing
 - D. Comparatoring
10. Ideally, the output voltage of an op-amp is zero when there is no input signal, however, in practical circuits, a small output voltage appears, this voltage is known as
 - A. minimum output voltage
 - B. pinch-off voltage
 - C. output offset voltage
 - D. saturation voltage
11. A Schmitt trigger is
 - A. a comparator with only one trigger point.
 - B. a comparator with hysteresis.
 - C. a comparator with three trigger points.
 - D. None of the above.
12. When the same signal is applied to both inverting and non-inverting input terminals of an ideal op-amp, the output voltage would be
 - A. zero (0) V
 - B. $+V_{\text{SAT}}$
 - C. $-V_{\text{SAT}}$
 - D. offset voltage
13. A differentiator is used to measure
 - A. the sum of the input voltages.
 - B. the difference between two voltages.
 - C. the area under a curve.
 - D. the rate of change of the input voltage.
14. The dc emitter current in each transistor of a differential amplifier equals
 - A. the tail current, I_T .
 - B. twice the tail current.
 - C. one-half the tail current.
 - D. zero.
15. The ratio of the differential gain and common gain of an op-amp
 - A. differential-common mode ratio
 - B. common-mode ratio
 - C. differential-mode rejection ratio
 - D. common-mode rejection ratio
16. An integrator circuit uses
 - A. a resistor in its feedback circuit.
 - B. an inductor in its feedback circuit.
 - C. a capacitor in its feedback circuit.
 - D. a resistor in its feedback circuit or uses a capacitor in its feedback circuit
17. The input impedance of an inverting amplifier is approximately equal to
 - A. R_i .
 - B. infinity.
 - C. zero.
 - D. R_F
18. Determine the cut-off frequency of an op-amp having specified values $B_1 = 1\text{ MHz}$ and $A_{\text{VD}} = 200\text{ V/mV}$.
 - A. 4 Hz
 - B. 1 kHz
 - C. 5 Hz
 - D. 1 MHz
19. Refer to the given figure. A square-wave input is applied to this amplifier. The output voltage is most likely to be



- A. a square wave.
B. a triangle wave.
C. a sine wave.
D. no output.
20. In an inverting amplifier, a virtual ground
- A. is no different from an ordinary ground.
B. can sink a lot of current.
C. usually has a significant voltage drop.
D. has the same potential as ground, yet it can sink no current.
21. For an op-amp having a slew rate of $SR = 2V/\mu s$, what is the maximum closed loop voltage gain that can be used when the input signal varies by 0.5 V in 10 μs ?
- A. 10 C. 20
B. 30 D. 40
22. The reason why a slight difference between the input bias current occurs in op-amps is due to the unsymmetrical circuit component parameters. This unsymmetrical condition also produces a difference in input voltage called what?
- A. Drift voltage
B. Differential voltage
C. Input offset voltage
D. Input threshold voltage
23. What is the maximum output voltage swing of an op-amp?
- A. +V to -V (supply voltage)
B. +VSAT to -VSAT
C. $+\frac{1}{2}V$ to $-\frac{1}{2}V$
D. depends on the input signal
24. Which of the following are variations of the basic summing amplifier?
- A. Averaging amplifier
B. Scaling amplifier
C. Both of the above
D. None of the above
25. A typical value of open-loop voltage gain for a 741 op-amp is
- A. 100 C. 0.5
B. 200,000 D. 100,000
26. Determine the common-mode gain of an op amp that has a differential voltage gain of 150×10^3 and a CMRR of 90 dB.
- A. 1.74 C. 2.74
B. 3.74 D. 4.74
27. The change in input offset current due to temperature change
- A. delta input offset current
B. slew rate
C. input offset current drift
D. PSRR
28. A good op-amp has a
- A. very high input resistance
B. very low input resistance
C. very high output resistance
D. very low CMRR
29. What do you call of the gain of an op-amp if operated in common mode input?
- A. differential gain
B. rejection gain
C. double-ended gain
D. common gain
30. The reduction of op-amps gain due to increasing operating frequency is called ____.
- A. cutoff
B. roll-off
C. diminishing factor
D. reduction step
31. What type of amplifier commonly used at the output stage of op-amps?
- A. differential amplifier
B. cascade-amplifier
C. complementary amplifier
D. darlington stage amplifier
32. An amplifier whose output is proportional to the difference between the voltages applied to its two inputs.
- A. differential amplifier
B. cascaded amplifier
C. current mirror
D. complementary amplifier
33. Ideally, op-amps have _____ output resistance.
- A. infinite
B. zero
C. variable
D. a highly stabilized
34. A voltage follower has a
- A. high input impedance.
B. low output impedance.
C. voltage gain of one.
D. All of the above.
35. If op-amps are operated in differential mode, its gain is technically termed as
- A. common-mode differential gain
B. differential gain
C. open-loop gain
D. closed-loop gain
36. Determine the output voltage of an op-amp for input voltages of $V_{i1} = 150 \mu V$ and $V_{i2} = 140 \mu V$. The amplifier has a differential gain of $A_d = 4000$ and the value of CMRR is 100.
- A. 32.6mV C. 45.8mV
B. 51.9mV D. 66.7mV
37. In an op-amp summing amplifier, the inputs are effectively isolated from each other because of the
- A. low output impedance of the op amp.
B. feedback resistor.
C. virtual ground.
D. None of the above.



38. When the inverting (-) input of an op amp is positive with respect to its noninverting (+) input, the output voltage is

- A. negative.
- B. positive.
- C. zero.
- D. None of the above.

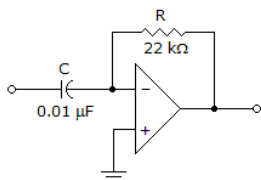
39. Op-amps have two input terminals namely, the inverting (-) and non-inverting (+) inputs. What is the significance of its name?

- A. If a sine-wave is applied to the inverting (-) input, the output will be inverted or shifted by 180° , while if applied to the non-inverting (+) there will be no phase shift at the output.
- B. If pulses are applied to the inverting (-) input, the positive pulse becomes negative at the output and vice versa, while if applied at the non-inverting (+) there will be no reversal of the pulse at the output.
- C. In dc amplifier applications, increasing input at the inverting (-) terminal causes the output to decrease and vice versa, while at the non-inverting (+) input, the output magnitude goes with the input.
- D. All of these are correct

40. When an op-amp circuit uses a single supply voltage, the dc output voltage from the op amp should be equal to

- A. $\frac{1}{2} V_{CC}$.
- B. zero.
- C. V_{CC} .
- D. $0.1 V_{CC}$.

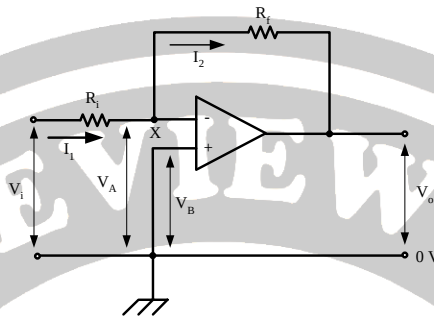
41. Refer to the given figure. This circuit is known as



- A. a noninverting amplifier.
- B. a differentiator.
- C. an integrator.

D. a summing amplifier.

42. In the inverting amplifier of the figure shown, $R_i = 1k\Omega$ and $R_f = 2k\Omega$. Determine the output voltage when the input voltage is 0.4 V.



- A. -0.80 V
- B. -0.10 V
- C. -0.90 V
- D. -0.11 V

43. Referring to the same figure in the above problem, what is the output when the input voltage is -1.2 V?

- A. +1.2 V
- B. +2.2 V
- C. +2.0 V
- D. +2.4 V

44. The current needed at the input of an op-amp to operate it normally

- A. input bias current
- B. input offset current
- C. input threshold current
- D. input holding current

45. The frequency at which the open-loop gain of an op-amp is 0.707 times its value at very low frequency

- A. threshold frequency
- B. break frequency
- C. minimum frequency
- D. operating frequency

46. Ideal op-amp requires no input current, but real op-amp needs a very small input current called input bias current. At both inputs, the bias currents have a slight difference. What do you call this difference?

- A. differential input current
- B. differential bias
- C. input offset difference
- D. input offset current

47. The operating mode of an op-amp, when both inputs are tied together or when the input signal is common to both inputs is called ____.

- A. differential mode
- B. rejection mode
- C. double-ended mode
- D. common mode

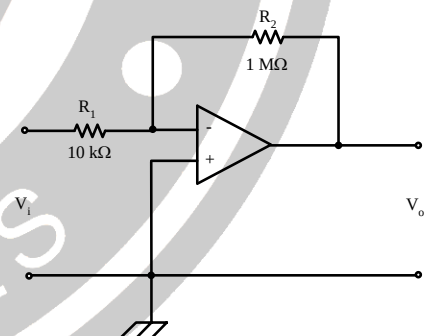
48. It is a very high-gain differential amplifier with very high input impedance and very low output impedance.

- A. par-amp
- B. op-amp
- C. differential amp
- D. complementary amp

49. An op-amp comparator that uses positive feedback is known as a

- A. zero-crossing detector.
- B. Schmitt trigger.
- C. peak detector.
- D. voltage follower.

50. The op amp shown below has an input bias current of 100 nA at 20°C . Calculate the output offset voltage due to the input bias current.



- A. 0.88 mV
- B. 0.77 mV
- C. 0.99 mV
- D. 0.11 mV

END